LLM Meta-Analysis Guide

Step-by-Step Guide: LLM Meta-Analysis Mini-Project

For students with no coding experience — no API keys, no programming, just free websites and AI chat tools

Total time needed: about 6–8 hours (you can split it over 2–3 days)

Before you start: what is a "meta-analysis"?

A meta-analysis is **not** five separate book reports stapled together. It's a report that **compares** papers and finds patterns across them.

Think of it like reviewing 4 restaurants. A summary says "Restaurant A serves pasta." A meta-analysis says "Three of the four restaurants focus on fresh ingredients, but they take completely different approaches to pricing — and all four struggle with slow service." **The comparison is the whole point.**

Nothing in this project requires coding, running models, or API keys. You will only need: a web browser, a free AI chat tool (Claude, ChatGPT, or Gemini), and Google Docs.

Step 1: Pick your theme (15 minutes)

All your papers must share **one topic**. Pick ONE of these beginner-friendly themes — each comes with ready-made papers you can use (links in Step 2):

Theme	In plain English	Difficulty
Instruction tuning	How raw models are taught to follow human instructions	⭐ Easiest
Efficient fine-tuning	How to adapt huge models cheaply (LoRA, QLoRA)	⭐⭐ Easy
Reasoning	How models are made to "think step by step"	⭐⭐ Easy
Alignment & safety	How models are trained to be helpful and harmless	⭐⭐⭐ Medium

Recommendation: if you're unsure, pick **Instruction tuning** — the papers are famous, well-explained online, and easy to compare.

Step 2: Get your papers (30 minutes)

You need 3–5 papers. **4 is the sweet spot** — enough to compare, not too much to read.

Option A — use a ready-made set (recommended)

All papers below are free on arXiv.org — click the link, then click "View PDF" on the left. No account needed.

Theme: Instruction tuning 1. InstructGPT — "Training language models to follow instructions with human feedback" (OpenAI, 2022) — https://arxiv.org/abs/2203.02155 2. FLAN — "Finetuned Language Models Are Zero-Shot Learners" (Google, 2021) — https://arxiv.org/abs/2109.01652 3. Self-Instruct — "Aligning Language Models with Self-Generated Instructions" (2022) — https://arxiv.org/abs/2212.10560 4. LIMA — "Less Is More for Alignment" (Meta, 2023) — https://arxiv.org/abs/2305.11206

Theme: Efficient fine-tuning 1. LoRA — "Low-Rank Adaptation of Large Language Models" (Microsoft, 2021) — https://arxiv.org/abs/2106.09685 2. QLoRA — "Efficient Finetuning of Quantized LLMs" (2023) — https://arxiv.org/abs/2305.14314 3. Prefix-Tuning — "Optimizing Continuous Prompts for Generation" (2021) — https://arxiv.org/abs/2101.00190 4. Mistral 7B (small-but-strong models) (2023) — https://arxiv.org/abs/2310.06825

Theme: Reasoning 1. Chain-of-Thought Prompting (Google, 2022) — https://arxiv.org/abs/2201.11903 2. Self-Consistency (Google, 2022) — https://arxiv.org/abs/2203.11171 3. Tree of Thoughts (2023) — https://arxiv.org/abs/2305.10601 4. DeepSeek-R1 — reasoning via reinforcement learning (2025) — https://arxiv.org/abs/2501.12948

Theme: Alignment & safety 1. InstructGPT (RLHF) (OpenAI, 2022) — https://arxiv.org/abs/2203.02155 2. Constitutional AI (Anthropic, 2022) — https://arxiv.org/abs/2212.08073 3. DPO — "Direct Preference Optimization" (2023) — https://arxiv.org/abs/2305.18290 4. LIMA — "Less Is More for Alignment" (Meta, 2023) — https://arxiv.org/abs/2305.11206

Option B — find your own papers

Go to **arxiv.org** or **scholar.google.com**, search your theme + "large language models", and pick papers that (a) are from 2021 or later,

(b) share your theme, and (c) have lots of citations (Google Scholar shows “Cited by ___” under each result — hundreds of citations = safe choice).

✅ **Checkpoint:** you have 3–5 PDF files downloaded, all on one theme.

Step 3: “Read” each paper with AI help (45–60 min per paper)

You do **not** need to understand every equation. Here’s the trick researchers actually use — read only these parts, in this order:

1. **Abstract** (the summary at the top) — what did they do?
2. **Introduction** — why does the problem matter?
3. **Figures and tables** — the results in picture form
4. **Conclusion / Discussion** — what did they find, what are the limits?

Skip the math-heavy “Method” details on first pass.

Use your AI chat tool as a reading tutor

Open Claude (or ChatGPT/Gemini), **upload the PDF** (paperclip/attach button — no API key needed, the normal chat works fine), and ask questions like:

> *“I’m a student with no coding background. Explain this paper to me in simple terms: What problem does it solve? What is their solution? What were the main results?”*

Then dig deeper with follow-ups:

> *“What datasets did they use, and what does each one contain?” “What model architecture did they use, and how big is it?” “What evaluation metrics did they use? Explain each metric like I’m new to this.” “What limitations do the authors admit? What did critics say about this paper?” “Explain Figure 1 to me.”*

⚠ **Two honesty rules (these protect your grade):** - **Verify** what the AI tells you by finding that claim in the actual paper. AI tools occasionally misremember numbers. If the AI says “they improved accuracy by 10%,” find that number in the paper’s tables. - **Never copy-paste AI text into your report.** Use the AI to *understand*, then close the chat and write from your own notes in your own words. Instructors can tell — and the assignment explicitly requires your own words.

Fill in this note template for EVERY paper

Copy this into a scratch doc and fill it per paper (short bullet answers are fine):

```
PAPER: (title, authors, year, where published)
LINK:
PROBLEM: What was broken or missing before this paper?
SOLUTION: What did they invent or do? (1-3 sentences, simple words)
MODEL: What model(s) did they use/build? How big?
DATA: What datasets did they train/test on?
METRICS: How did they measure success? (benchmark names + what they test)
RESULTS: The 2-3 most impressive numbers or findings
LIMITATIONS: What do the authors admit doesn't work well?
MY TAKE: One sentence — what surprised me or seems important?
```

✅ **Checkpoint:** one filled template per paper. This is 80% of the hard work — the report almost writes itself from here.

Step 4: Build the comparison table (30 minutes)

Make one table with a row per aspect and a column per paper. Example skeleton:

	Paper 1	**Paper 2**	**Paper 3**	**Paper 4**
Goal				
Approach				
Model & size				
Data				
Evaluation				
Key result				

Main limitation				
Reproducible? (code/data public?)				

Fill it straight from your Step 3 templates. When you're done, stare at the table and ask: *what do the columns have in common? Where do they disagree?* Write those observations down — they are your Section 3 and Section 4 content.

Step 5: Write the report (2–3 hours)

Open a **new Google Doc** at docs.google.com (free Google account). One single document, **no tabs** (don't use the "tabs" feature in the left sidebar — everything in one continuous file). Max **6 pages**.

Write the sections in this order (it's easier than top-to-bottom):

First → Section 2: Paper Summaries (½ page each). For each paper: full citation (see format below), then problem → solution → results → data/model/metrics. Straight from your templates.

Second → Section 3: Comparative Analysis (1–1.5 pages). Paste in your table, then write paragraphs about the differences and similarities it reveals. Cover: objectives, architectures/innovations, training strategies, benchmarks, strengths/limitations/reproducibility.

Third → Section 4: Insights & Reflection (1 page). Answer the four questions from the assignment: What trends appear across all papers? Which approach seems most promising, and why? What limitations do multiple papers share? What should researchers do next?

Fourth → Section 1: Introduction (½ page). Now that you know what you found, the intro is easy: what LLMs are (2–3 sentences), your theme and why it matters, and the list of your papers with sources.

Last → Section 5: Conclusion (½ page) + a **paper list with clickable links** at the end (required!). To make a link clickable in Google Docs: paste the URL and press space, or select text and press Ctrl+K (Cmd+K on Mac).

Citation format (simple APA-style is fine): > Ouyang, L., et al. (2022). *Training language models to follow instructions with human feedback.* arXiv:2203.02155.

💡 Good AI uses at this stage: "check my grammar," "is my comparison of X and Y factually accurate?", "suggest a clearer way to phrase this sentence." Bad AI use: "write my Section 4 for me."

Step 6: Make the Google Doc PUBLIC and submit (10 minutes)

This step loses people points every year. Do it exactly like this:

1. In your Google Doc, click the blue **Share** button (top-right).
2. Under **General access**, change "Restricted" to **"Anyone with the link."**
3. Leave the role as **Viewer**.
4. Click **Copy link**, then **Done**.
5. **Test it:** open an incognito/private browser window (Ctrl+Shift+N in Chrome), paste the link. If you can read the doc **without logging in**, it works. If it asks you to sign in or request access — go back to step 1.
6. Paste the tested link into the DI Learning platform.

Optional: save to GitHub (no coding needed)

In Google Docs: File → Download → **PDF Document (.pdf)**, name it `meta_analysis_llms.pdf`. Then on github.com, open your repository → **Add file → Upload files** → drag the PDF in → type a message like "Add meta-analysis report" → **Commit changes**. That's it — uploading a file on GitHub's website is just drag-and-drop.

Final checklist before submitting

- [ ] 3–5 papers, all on ONE shared theme
- [ ] All 5 sections present (Intro, Summaries, Comparison, Insights, Conclusion)
- [ ] At least one comparison table
- [ ] Report is 6 pages or less
- [ ] Written in YOUR words (AI used for understanding and polishing only)
- [ ] Full citations + clickable links to every paper
- [ ] One single Google Doc, no tabs
- [ ] Link tested in incognito mode — opens without login
- [ ] Link submitted on the DI Learning platform

Common mistakes to avoid

1. **Summarizing instead of comparing.** Sections 3–4 are worth the most — that's where the "meta" in meta-analysis lives.
2. **Papers with no common theme.** "Four random LLM papers" is not a theme.
3. **Copy-pasting AI output.** Use AI as a tutor, not a ghostwriter.
4. **Forgetting to test the public link.** Always check in incognito.
5. **Going over 6 pages.** If too long, shrink the summaries — never cut the comparison or insights.

Good luck — you've got this! 🍀